

Mystery Meat Analysis

BIOLM0051

Archie Benn

Scenario

A suitcase containing several packages of unidentified frozen meat has been intercepted at Heathrow Airport by UK Border Force officers during routine inspections. The shipment had no import documentation, raising concerns that the meat may originate from protected or endangered species.

Given the serious implications for public health, food safety, and wildlife conservation, the case has been referred to the Animal and Plant Health Agency (APHA). As a bioinformatician assisting with the inquiry, you have been tasked with determining the species of origin for each meat sample. DNA barcoding has already been performed in the laboratory, and you have now received the resulting FASTAQ sequence files for analysis.

Abstract

Introduction

The Washington Convention, also known as CITES, was established in 1975 a key legal framework to protect endangered species from global wildlife, yet from 1975 to 2014 the volume of CITES-listed organisms traded per year quadrupled to over 100 million whole-organism equivalents [1]. This

Methods
Pipeline Overview
Results

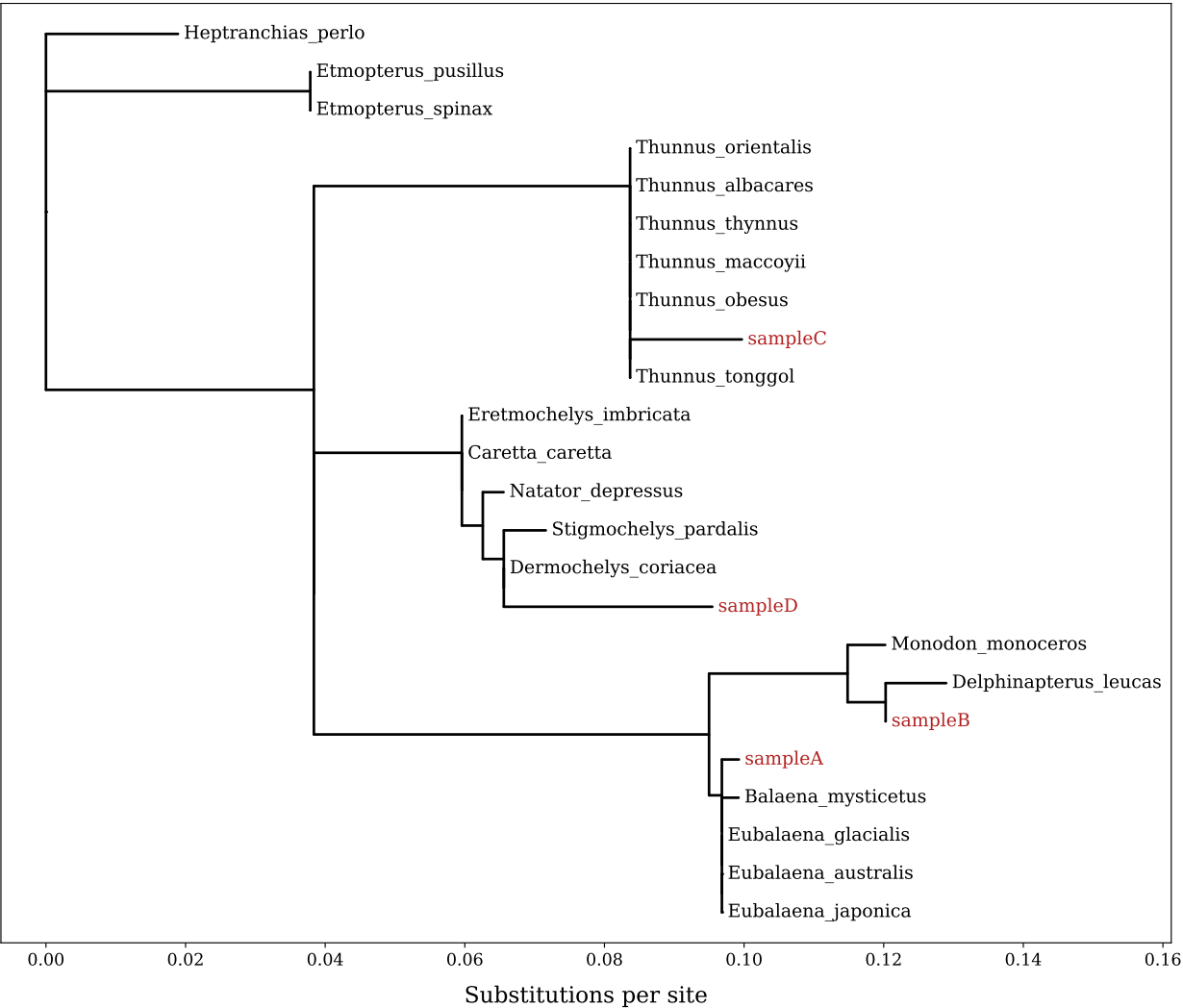


Figure 1: Maximum-likelihood phylogenetic tree with placement of unknown meat samples and their top BLAST hits

Discussion

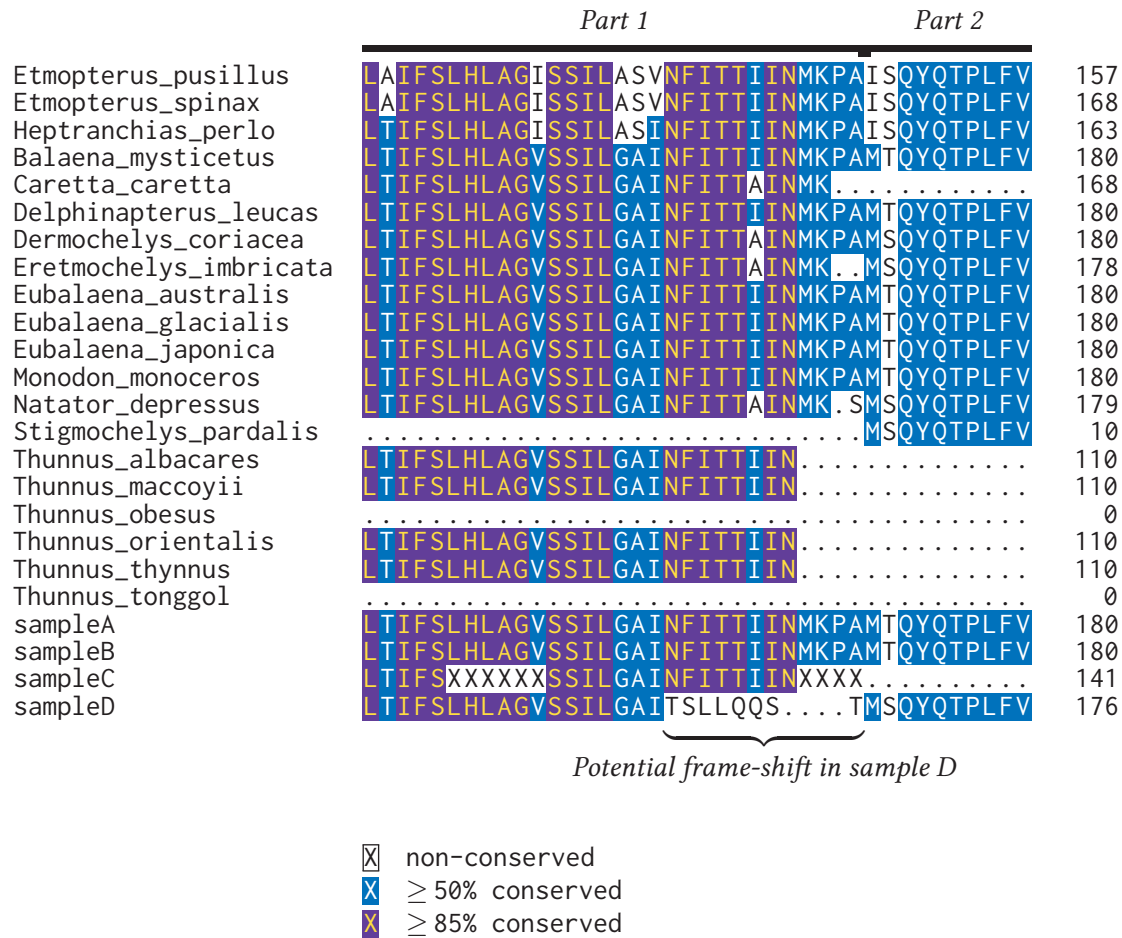


Figure 2: Window of supermatrix alignment with potential frame-shift error in sampleD near the highly-conserved NFITTIIN region.

Dermochelys_c	441	CACCTAGCTGGTGTTCATCAATTTTAGGAGCTATTAACCTCATTACTAC	490
sampleD_part1	451	CACCTAGCTGGTGTTCATCAATTTTAGGAGCTATT-ACTCATTACTAC	499



Figure 3: Needleman-Wunsch global pairwise alignment at the tail end of sampleD_part1 against closest reference sequence (*Dermochelys coriacea*). The alignment gap indicates a possible base miss at position 486 in sampleD_part1 during sequencing.

References

- [1] Michael Harfoot, Satu AM Glaser, Derek P Tittensor, Gregory L Britten, Claire McLardy, Kelly Malsch, and Neil D Burgess. Unveiling the patterns and trends in 40 years of global trade in cites-listed wildlife. *Biological Conservation*, 223:47–57, 2018.

Supplementary Data

Supplementary data and all scripts can be found within the GitHub repository for this project: https://github.com/archiebenn/BIOLM0051_analysis.git

Appendix: Exploratory Figures for Manual Correction of Suspected Sequencing Error

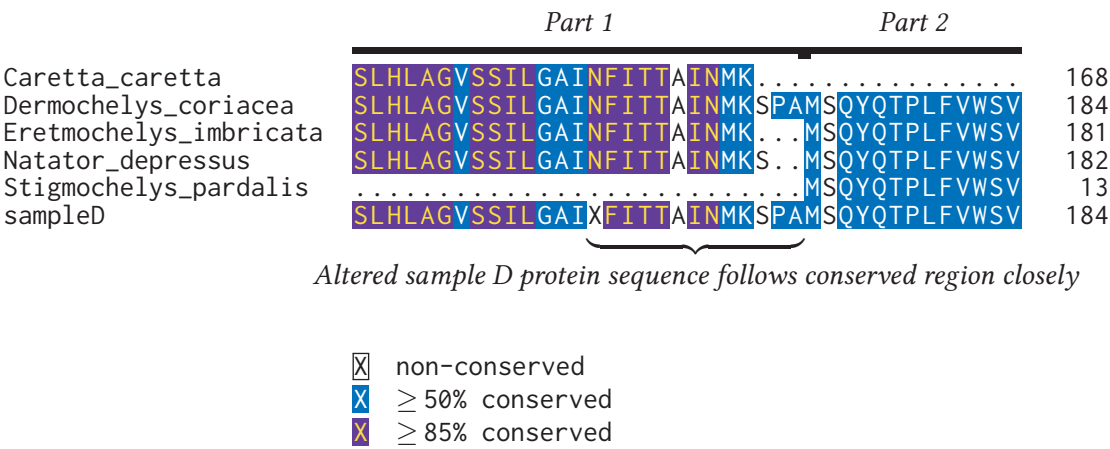


Figure 4: Subset of supermatrix alignment with sampleD and inner clad after inserting one ambiguous base (N) to the nucleotide sequence of sampleD.part1. This base resolves the apparent frame-shift, resulting in a protein sequence which aligns much more closely to highly-conserved NFITTIIN region compared to in Figure 2.

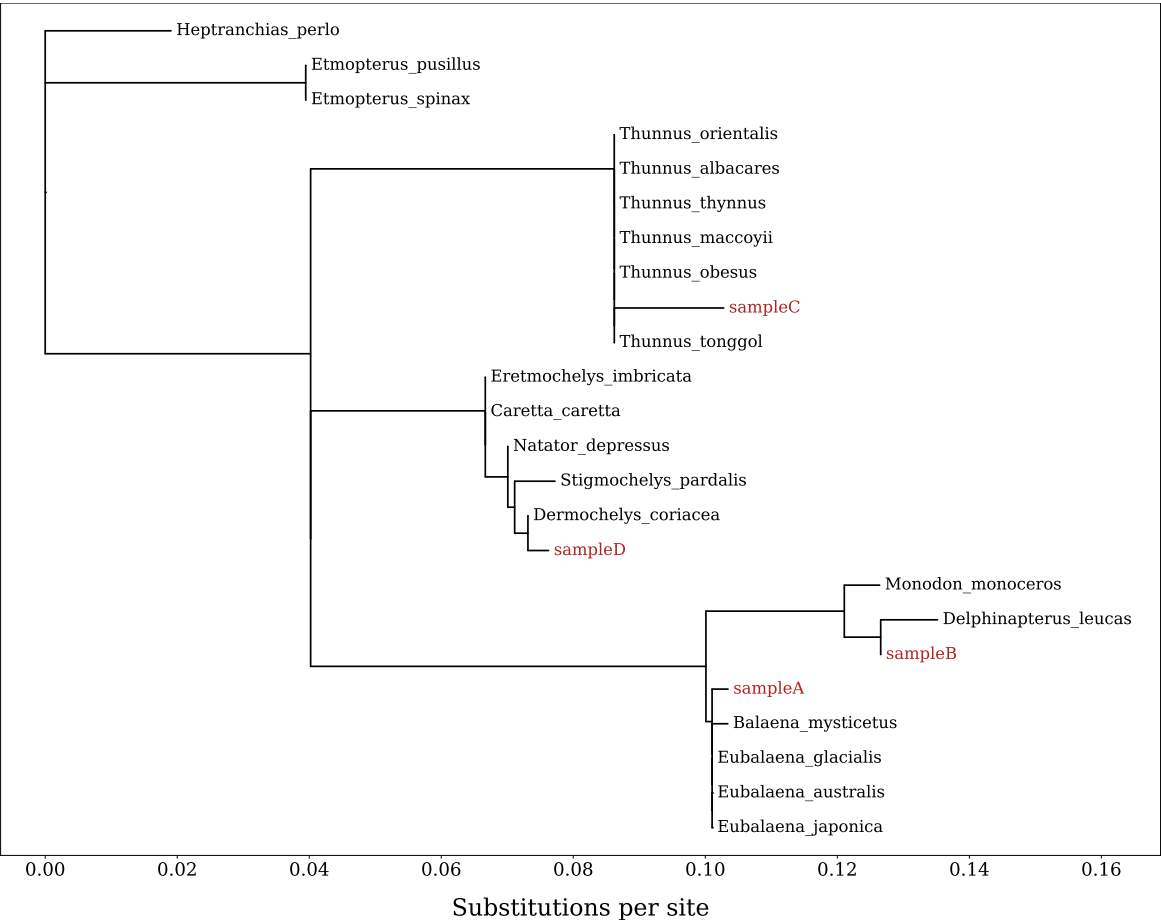


Figure 5: Maximum-likelihood phylogenetic tree after manually corrected alignment in which one ambiguous base (N) was inserted into sampleD.part1. This tree is shown to demonstrate the impact of sequencing error and downstream frame-shift on branch lengths compared to Figure 1.